

Dimix: A Cross-Dimensional Mixed Reality System Based on Latent Diffusion Model

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Figure 1: The real world seamlessly blends with a world generated by Latent Diffusion Model

ABSTRACT

Dimix is a system that integrates 2D images generated by Latent Diffusion Models (LDMs) [et al. 2022] with 3D and interactive Mixed Reality (MR) experiences. This system extracts regions from MR scenes based on panoramic images, adds optional mask images, and applies image-inpainting using LDMs to output an LDMs-generated image (LDMs image) that naturally extends the scene. Furthermore, by performing depth estimation on the images and reconstructing them as 3D scenes, the LDMs image can be treated in the same manner as 3D objects. This enables both basic features like occlusion and collision detection and highly interactive operations such as ink painting, achieving high immersion and realism. Additionally, the system incorporates real-time object detection to constrain the inpainting area, making the LDMs image more convincing. All processing is performed in real-time, allowing users to interact with the world in 3D without waiting for loading or any other preparation by simply uploading their preferred panorama image to the application.

To our best knowledge, Dimix is the first system to seamlessly integrate LDMs images into interactive and three-dimensional MR

experiences. Users can immerse themselves in an unprecedented space where the real world seamlessly blends with a world generated by a neural network, offering a glimpse into the future of MR experiences.

ACM Reference Format:

Daiki Taniguchi. 2023. Dimix: A Cross-Dimensional Mixed Reality System Based on Latent Diffusion Model. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Immersive Pavilion (SIGGRAPH '23 Immersive Pavilion)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3588027.3595592>

1 MIXED REALITY WITH LDMS

Dimix extracts a designated region (e.g., 512x512 pixels) from an equirectangular panoramic image, and sends it along with an arbitrary mask image and prompt to a server. At the server, a generative model based on Stable Diffusion [et al. 2022] inpaints the image and sends it back to the device. The device can then seamlessly integrate the transformed image into the original space by replacing the designated region of the panoramic image in real-time. With recent advancements in LDMs, various worlds can be generated by providing various prompts. Figure 2 shows an example of the world before and after transformation by Dimix.

The extraction of the transformation region and replacement of the transformed image typically involves a copy of the image from the GPU to the CPU, which pauses the application for a moment. However, by performing all processes asynchronously, our system can perform image replacements without any interruption.

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SIGGRAPH '23 Immersive Pavilion, August 06–10, 2023, Los Angeles, CA, USA

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ACM ISBN 979-8-4007-0151-1/23/08.

<https://doi.org/10.1145/3588027.3595592>

2 3D RECONSTRUCTION FROM LDMS IMAGE

Our system provides a technique for handling panoramic images and LDMS images in a 3D manner. Specifically, we perform depth estimation alongside the LDMS image and then reconstruct the space in 3D from the depth map in real-time. The depth estimation can be performed with high accuracy by using Pano3d [et al. 2021], a state-of-the-art method for depth estimation from spherical panoramas. The reconstruction of the 3D space is performed by applying our previous research, the Garage system [Taniguchi 2021], a GPU particle-based technology that reconstructs real space with high realism and interactivity. The reconstructed space can be handled as 3D, allowing for collisions and occlusions, dramatically enhancing the persuasiveness of LDMS images. Figure 4 shows an example of collisions and occlusions with 3D objects.



Figure 2: (a) Original Room (b) Deep Forest (c) Japanese Sakura (d) Space Opera

3 INTERACTION WITH LDMS IMAGE

Additionally, Dimix implements grid-based spatial partitioning [Green 2010] to efficiently search for neighboring GPU particles, enabling fast detection of mutual interference between particles. This allows for complex interactions within the 3-dimensional reconstructed LDMS world. For example, users can splash paint in their preferred color and paint the world as they like (Figure 3). This process involves a physics simulation using a diffusion equation and our own constraints after detecting collisions in 3D and converting the collision area to a 2D equirectangular coordinate system, writing ink splash areas read out via GBuffer. All processing is done on the GPU, achieving a very fast and convincing expression including ink drips. The experience of freely painting a world that changes dynamically and beautifully with LDMS is a very attractive experience that cannot be experienced in reality. There are also other attractive experiences, such as opening portal gates to change space or voxelizing space altered by generative models.

4 HIGH QUALITY INPAINTING

If random regions are selected for transformation, unintended results may occur. For instance, if regions such as floors or ceilings that lack any features are chosen, the transformation may not take place at all, or the results may be distorted or unnatural. To address this issue, we prioritize regions that contain objects detected



Figure 3: High quality ink painting

by an object detection algorithm, resulting in more realistic transformations. Additionally, the object detection algorithm runs on the device, rather than sending data to a server, enabling fast and delay-free detection that adapts to the user's viewpoint. Figure 4 compares our high-quality conversions with failed ones due to random selection, demonstrating the effectiveness of our approach.



Figure 4: (a) Failed translation with random regions (b)(c) Succeed translation with object detection

5 CONCLUSION

While most VR/AR applications that incorporate LDMS are limited to passive viewing experiences, Dimix sets itself apart by enabling real-time interaction with the LDMS-Generated World in 3D. The advancements in LDMS are impressive and hold great potential for creating future experiences. However, the current lack of temporal continuity and interactivity in LDMS poses a challenge for integrating them into xR experiences, which include virtual, augmented, and mixed reality. By combining our established 3D and AR technologies with new LDMS technologies, we have been able to realize next-generation MR experiences, as demonstrated by Dimix.

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